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Report
On

"ECG Signal Analysis Using MATLAB"

Signals and Systems (BEC403)

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1. Abstract

Electrocardiogram (ECG) signals represent the electrical activity of the human heart and are widely used in medical diagnostics. Accurate analysis of ECG signals helps in identifying heart conditions such as arrhythmia, tachycardia, and bradycardia.

This project presents the analysis of ECG signals using MATLAB. The ECG data is obtained from the MIT-BIH Arrhythmia Database. The raw ECG signal is processed using signal processing techniques including filtering, peak detection, and feature extraction.

A bandpass filter is applied to remove noise and baseline drift from the ECG signal. R-peaks are detected from the filtered ECG waveform, which are used to calculate RR intervals and heart rate. Heart rate variability (HRV) and peak amplitude features are also calculated.

Finally, the ECG condition is evaluated based on the calculated parameters and displayed through graphical visualization using MATLAB.

2. Introduction

Electrocardiography is a widely used technique for monitoring the electrical activity of the heart. The ECG signal provides valuable information about heart rhythm and cardiovascular health.

A normal ECG waveform consists of several components:

- **P wave** – represents atrial depolarization
- **QRS complex** – represents ventricular depolarization
- **T wave** – represents ventricular repolarization

Among these components, the **R-peak** is the most prominent feature in an ECG signal and is commonly used for calculating heart rate.

However, ECG signals often contain noise caused by:

- Power line interference
- Muscle activity
- Motion artifacts

- Baseline drift

Signal processing techniques such as filtering and peak detection are required to remove noise and extract meaningful features.

MATLAB provides powerful tools for signal processing and visualization, making it suitable for ECG signal analysis.

3. Objectives

The main objectives of this project are:

1. To load ECG signal data from the MIT-BIH dataset.
2. To preprocess the ECG signal using filtering techniques.
3. To detect R-peaks from the ECG waveform.
4. To calculate RR intervals between successive heartbeats.
5. To determine heart rate and heart rate variability.
6. To visualize ECG signal characteristics using MATLAB plots.
7. To generate a simple diagnostic interpretation based on calculated parameters.

4. Dataset Description

The ECG signal used in this project is obtained from the MIT-BIH Arrhythmia Database, which is a widely used dataset for cardiovascular signal analysis.

Characteristics of the dataset include:

1. Sampling frequency: 360 Hz
2. Each ECG record contains two channels of ECG signals recorded over time.
3. In this project, the ECG signal is read from a .dat file and the first channel is selected for analysis. The first 10 seconds of ECG data are extracted and used for signal processing.

5. Methodology

The ECG signal analysis is carried out through the following steps.

5.1 Loading ECG Data

The ECG data is loaded from the dataset file using MATLAB file operations. The signal is stored in an array for further processing.

5.2 Signal Segmentation

To simplify the analysis, the first 10 seconds of the ECG signal are selected. The time axis is generated using the sampling frequency.

5.3 Noise Filtering

ECG signals often contain unwanted noise. To remove noise, a bandpass Butterworth filter is applied.

Filter range:

0.5 Hz – 40 Hz

This filter removes baseline drift and high frequency noise while preserving important ECG features.

5.4 R-Peak Detection

R-peaks are detected using MATLAB's findpeaks() function.

The function identifies local maxima in the filtered ECG signal using parameters such as:

- Minimum peak height
- Minimum peak distance

These parameters help avoid false peak detection.

5.5 RR Interval Calculation

The RR interval is the time difference between two consecutive R-peaks.

RR intervals are calculated by finding the difference between successive peak locations.

5.6 Heart Rate Calculation

Heart rate is calculated using the formula:

Heart Rate (BPM) = 60 / RR Interval

The average heart rate is then computed from all detected heartbeats.

5.7 Heart Rate Variability

Heart Rate Variability (HRV) represents the variation in time intervals between heartbeats.

HRV is calculated as the standard deviation of RR intervals.

5.8 ECG Diagnosis

The ECG condition is classified based on heart rate and HRV values.

Condition	Criteria
Bradycardia	Heart Rate < 60 BPM
Tachycardia	Heart Rate > 100 BPM
Possible Arrhythmia	HRV > 0.25
Normal ECG	Otherwise

6. MATLAB Implementation

1. Load ECG data from dataset
2. Extract a segment of ECG signal
3. Apply bandpass filtering
4. Detect R-peaks
5. Calculate RR intervals
6. Compute heart rate and HRV
7. Display graphical results

Section 6.1 MATLAB Program

```
clc
clear
close all

%% PARAMETERS
fs = 360;
filename = 'data/mit-bih-arrhythmia-database-1.0.0/234.dat';

%% LOAD ECG SIGNAL
fid = fopen(filename,'r');
data = fread(fid,[2, inf],'int16');
fclose(fid);

ecg = data(1,:);

%% SELECT FIRST 10 SECONDS
duration = 10;
samples = fs * duration;

ecg_segment = ecg(1:samples);
t = (0:samples-1)/fs;

%% FILTER ECG (0.5 - 40 Hz)
low = 0.5/(fs/2);
high = 40/(fs/2);

[b,a] = butter(4,[low high],'bandpass');
filtered_ecg = filtfilt(b,a,ecg_segment);

%% R-PEAK DETECTION
[peaks,locs] = findpeaks(filtered_ecg,...
    'MinPeakHeight',1000,...
    'MinPeakDistance',0.6*fs);

%% RR INTERVAL
RR = diff(locs)/fs;

%% HEART RATE
HR = 60./RR;
avgHR = mean(HR);

%% HRV
HRV = std(RR);

%% PEAK AMPLITUDE FEATURES
meanPeak = mean(peaks);
maxPeak = max(peaks);
minPeak = min(peaks);

%% DIAGNOSIS
if avgHR < 60
    status = "Bradycardia";
elseif avgHR > 100
    status = "Tachycardia";
elseif HRV > 0.25
    status = "Possible Arrhythmia";
else
    status = "Normal ECG";
end

%% PRINT RESULTS
fprintf("\n------ ECG REPORT ------\n")
fprintf("Average HR: %.2f BPM\n",avgHR)
fprintf("HRV: %.3f sec\n",HRV)
fprintf("Mean RR Interval: %.3f sec\n",mean(RR))
```

```
fprintf("Mean R Peak Amplitude: %.2f\n",meanPeak)
fprintf("Max R Peak Amplitude: %.2f\n",maxPeak)
fprintf("Min R Peak Amplitude: %.2f\n",minPeak)
fprintf("Diagnosis: %s\n",status)
fprintf("------------------------\n")

%% PLOTS (ALL IN ONE WINDOW)

figure('Name','ECG Analysis Dashboard','NumberTitle','off')

% Raw ECG
subplot(3,2,1)
plot(t,ecg_segment)
title('Raw ECG Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

% Filtered ECG
subplot(3,2,2)
plot(t,filtered_ecg)
title('Filtered ECG')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

% R Peak Detection
subplot(3,2,3)
plot(t,filtered_ecg)
hold on
plot(locs/fs,peaks,'ro')
title('Detected R Peaks')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

% RR Intervals
subplot(3,2,4)
plot(RR,'o-')
title('RR Intervals')
xlabel('Beat Number')
ylabel('Seconds')
grid on

% Heart Rate Variation
subplot(3,2,5)
plot(HR,'o-')
title('Heart Rate Variation')
xlabel('Beat Number')
ylabel('BPM')
grid on

% Text summary
subplot(3,2,6)
axis off
text(0.1,0.7,['Heart Rate: ' num2str(avgHR,'%.2f') ' BPM'],'FontSize',12)
text(0.1,0.5,['HRV: ' num2str(HRV,'%.3f') ' sec'],'FontSize',12)
text(0.1,0.3,['Diagnosis: ' char(status)],'FontSize',12)
```

Key MATLAB functions used in the implementation include:

Function	Purpose
fread()	Read ECG data file

Function	Purpose
butter()	Design Butterworth filter
filtfilt()	Apply zero-phase filtering
findpeaks()	Detect R peaks
diff()	Calculate RR intervals
mean()	Calculate average heart rate
std()	Calculate HRV

7. Results and Analysis

The MATLAB program generates several plots to visualize the ECG signal characteristics.

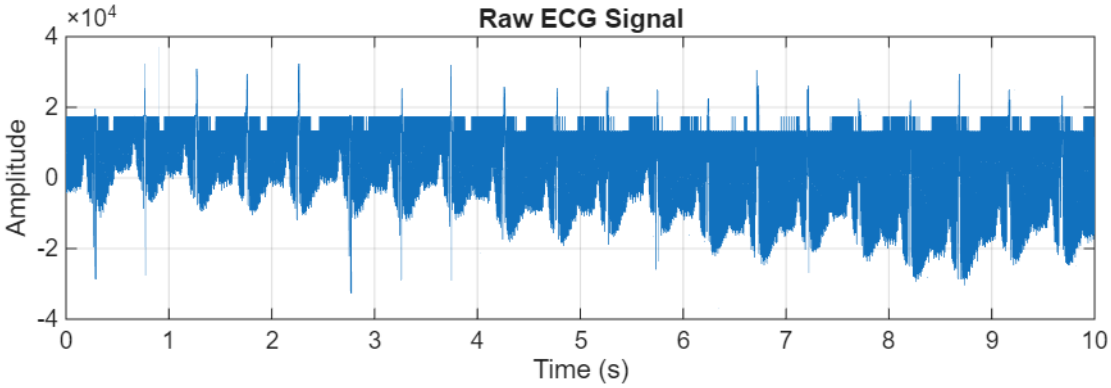

Figure 1: Raw ECG Signal

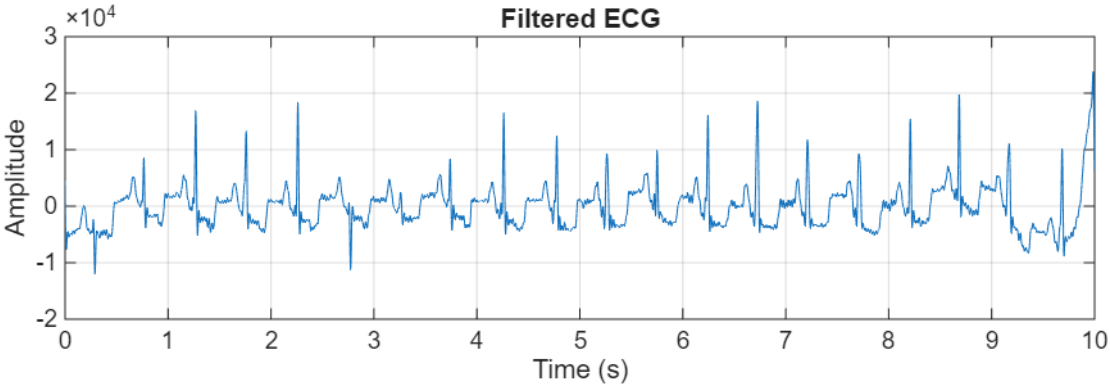

Figure 2: Filtered ECG Signal

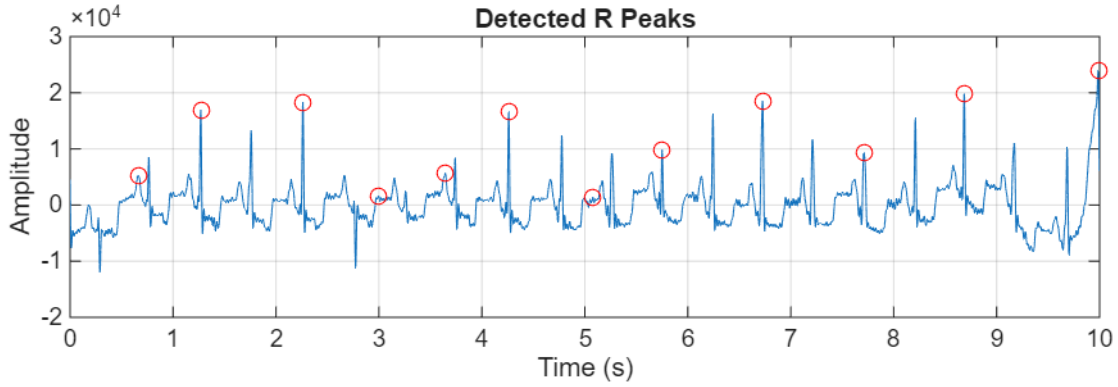

Figure 3: Detected R Peaks

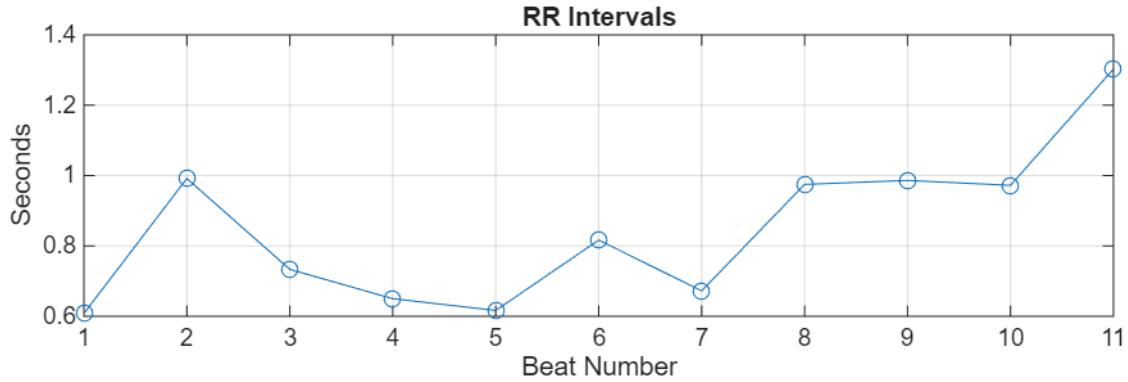

Figure 4: RR Intervals

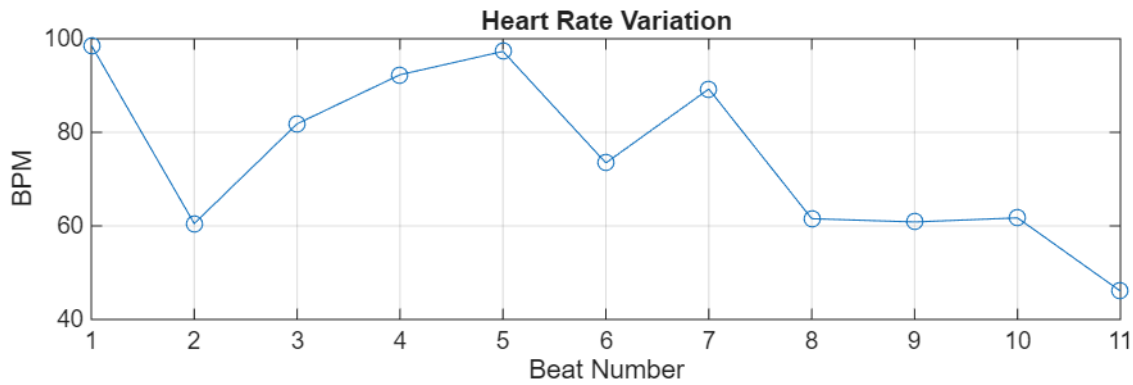

Figure 5: Heart Rate Variation

Heart Rate: 74.86 BPM

HRV: 0.218 sec

Diagnosis: Normal ECG

Figure 6: ECG Summary

8. Observations

From the analysis of the ECG signal, the following values were obtained:

Average Heart Rate ≈ **74.86 BPM**

Heart Rate Variability ≈ **0.218 sec**

Mean RR Interval ≈ **0.848 sec**

The heart rate lies within the normal range of 60–100 BPM.

Therefore the ECG signal is classified as:

Normal ECG

9. Conclusion

In this project, ECG signal analysis was successfully implemented using MATLAB. The ECG signal was filtered using a bandpass filter to remove noise and baseline drift.

R-peaks were detected from the filtered ECG signal, which allowed the calculation of RR intervals, heart rate, and heart rate variability. These parameters were used to determine the condition of the ECG signal.

The results were visualized through multiple plots, providing a clear understanding of the ECG signal characteristics.

This project demonstrates the application of signal processing techniques in biomedical signal analysis using MATLAB.

The results demonstrate how signal processing techniques can be applied effectively to biomedical signals for heart monitoring and diagnosis.

10. References

1. **Source:**
   PhysioNet MIT-BIH Arrhythmia Database
   https://physionet.org/content/mitdb/
2. MATLAB Signal Processing Toolbox Documentation
3. Biomedical Signal Processing – Rangayyan